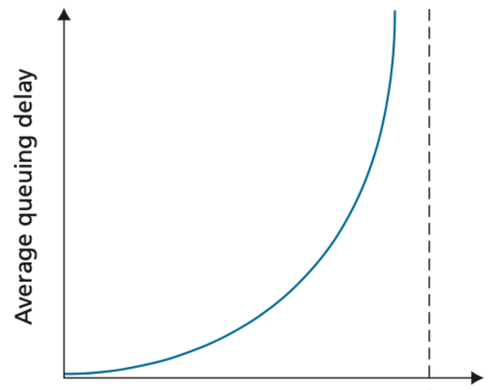
INTERACTIVE END-OF-CHAPTER EXERCISES

QUEUING DELAY

Consider the queuing delay in a router buffer, where the packet experiences a delay as it waits to be transmitted onto the link. The length of the queuing delay of a specific packet will depend on the number of earlier-arriving packets that are queued and waiting for transmission onto the link. If the queue is empty and no other packet is currently being transmitted, then our packet's queuing delay will be zero. On the other hand, if the traffic is heavy and many other packets are also waiting to be transmitted, the queuing delay will be long.



Assume a constant transmission rate of R = 400000 bps, a constant packet-length L = 3700 bits, and a is the average rate of packets/second. Traffic intensity I = La/R, and the queuing delay is calculated as I(L/R)(1 - I) for I < 1.

QUESTION LIST

1. In practice, does the queuing delay tend to vary a lot? Answer with Yes or No

2. Assuming that a = 33, what is the queuing delay? Give your answer in milliseconds (ms)

3. Assuming that a = 57, what is the queuing delay? Give your answer in milliseconds (ms)

4. Assuming the router's buffer is infinite, the queuing delay is 2.3056 ms, and 1766 packets arrive. How many packets will be in the buffer 1 second later?

5. If the buffer has a maximum size of 748 packets, how many of the 1766 packets would be dropped upon arrival from the previous question?

SOLUTION

1. Yes, in practice, queuing delay can vary significantly. We use the above formulas as a way to give a rough estimate, but in a real-life scenario it is much more complicated.

2. Queuing Delay = I(L/R)(1 - I) * 1000 = 0.3053*(3700/400000)*(1-0.3053) * 1000 = 1.9619 ms.

3. Queuing Delay = I(L/R)(1 - I) * 1000 = 0.5273*(3700/400000)*(1-0.5273) * 1000 = 2.3056 ms.

4. Packets left in buffer = a - floor(1000/delay) = 1766 - floor(1000/2.3056) = 1333 packets.

5. Packets dropped = packets - buffer size = 1766 - 748 = 1018 dropped packets.

QUESTION 1 OF 5

In practice, does the queuing delay tend to vary a lot? Answer with Yes or No

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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